

Olist E-Commerce Data Analysis

Project Overview

This project analyzes an e-commerce dataset using Python to understand sales performance, customer behavior, product categories, sellers, and delivery performance.

The main goal of the project is to explore the raw data, identify important patterns, answer business-related questions, and turn the results into meaningful business insights.

Dataset

The project uses the Olist Brazilian E-Commerce dataset.

The dataset contains information about:

- Orders
- Customers
- Order items
- Products
- Sellers
- Prices
- Freight
- Order status
- Delivery dates
- Customer locations
- Product categories

Several related datasets were combined to create a merged dataset for the analysis.

Tools Used

- Python
- Pandas
- Jupyter Notebook
- Microsoft Word for the analysis report

Data Analysis Process

The analysis was completed in several steps:

1. Loaded the datasets using Python.
2. Explored the structure and dimensions of the data.
3. Examined column names and data types.
4. Checked missing values.
5. Combined related datasets using common identifiers.
6. Performed exploratory data analysis.

7. Created business questions based on the available data.
8. Calculated different sales, customer, and delivery metrics.
9. Interpreted the results using key findings.
10. Documented the analysis and conclusions in a Word report.

Business Questions

The analysis examined different aspects of the business, including:

- Which product categories generate the highest revenue?
- Which customer states generate the highest revenue?
- Which sellers generate the highest revenue?
- Which months generate the highest revenue?
- What is the distribution of order statuses?
- How many unique customers placed orders?
- How many customers made more than one purchase?
- What is the average order value?
- What is the delivery performance compared with the estimated delivery date?
- Which customer states have the longest average delivery time?
- Which product categories are most frequently ordered in major customer states?
- How does revenue change over time?

Additional analyses were also performed to explore the dataset from different business perspectives.

Key Findings

The analysis showed several important patterns in the Olist dataset.

Some product categories generated considerably more revenue than others, while certain customer states contributed a much larger share of total revenue.

The analysis also showed that most orders had a `delivered` status. Customer analysis revealed that the majority of customers made only one purchase, while a smaller percentage of customers placed more than one order.

The average order value was approximately 137.75, and delivery-date analysis showed that orders were delivered, on average, before the estimated delivery date.

The analysis also identified differences in average delivery time between customer states and differences in revenue across months, product categories, sellers, and geographic regions.

Conclusion

Overall, this project provided practical experience in using Python and Pandas for exploratory data analysis. It demonstrated how raw e-commerce data can be cleaned, combined, analyzed, and interpreted to answer business questions.

The project also helped develop an understanding of how data analysis can be used to identify patterns in sales, customers, products, sellers, and delivery performance.

Project Files

- `Olist_Analysis.py` — Python analysis code
- `Olist_Analysis_Report.docx` — Detailed analysis report and key findings
- `README.md` — Project documentation